

Diversion Ratios

Chris Conlon

Fall 2026

NYU Stern

What are Diversion Ratios?

Horizontal Merger Guidelines (2010 rev.)

*In some cases, the Agencies may seek to quantify the extent of direct competition between a product sold by one merging firm and a second product sold by the other merging firm by estimating the diversion ratio from the first product to the second product. The diversion ratio is the **fraction of unit sales lost by the first product due to an increase in its price that would be diverted to the second product**. Diversion ratios between products sold by one merging firm and products sold by the other merging firm can be very informative for assessing unilateral price effects, with **higher diversion ratios indicating a greater likelihood of such effects**. Diversion ratios between products sold by merging firms and those sold by **non-merging firms have at most secondary predictive value**.*

Raise price of good j . People leave. What fraction of leavers switch to k ?

$$D_{jk}(p_j, \mathbf{p}_{-j}) = \frac{\frac{\partial q_k}{\partial p_j}}{\left| \frac{\partial q_j}{\partial p_j} \right|}$$

It's one of the best ways economists have to characterize competition among sellers.

- ▶ High Diversion: Close Substitutes $\rightarrow$ Mergers more likely to increase prices.
- ▶ Very low diversion $\rightarrow$ products may not be in the same market.
(ie: Katz & Shapiro). This is just hypothetical monopolist or SSNIP test.
- ▶ Demand Derivatives NOT elasticities.
- ▶ No equilibrium responses.

Nevo (2000) Example

#	Brand	K Corn Flakes	K Raisin Bran	K Frosted Flakes	K Rice Krispies
1	K Corn Flakes	-3.696	.023	.500	.010
2	K Raisin Bran	.023	-2.061	.088	.051
3	K Frosted Flakes	.361	.059	-3.546	.028
4	K Rice Krispies	.010	.048	.040	-1.320
5	K Frosted Mini Wheats	.000	.053	.003	.057
6	K Froot Loops	.000	.010	.008	.038
7	K Special K	.155	.072	.248	.039
8	K NutriGrain	.270	.094	.313	.023
9	K Crispix	.003	.038	.020	.079
10	K Cracklin Oat Bran	.000	.023	.001	.046
11	GM Cheerios	.007	.080	.035	.069
12	GM Honey Nut Cheerios	.001	.017	.017	.043
13	GM Wheaties	.503	.113	.445	.029
14	GM Total	.140	.064	.238	.042
15	GM Lucky Charms	.000	.012	.010	.041
16	GM Trix	.000	.010	.009	.043
17	GM Raisin Nut	.007	.137	.043	.059
18	P Raisin Bran	.014	.232	.063	.050
19	P Grape Nuts	.001	.048	.006	.050
20	Q 100% Natural	.000	.023	.002	.048
21	Q Life	.003	.038	.052	.048
22	Q CapNCrunch	.001	.013	.015	.038
23	R Chex	.005	.037	.028	.081
24	N Shredded Wheat	.002	.081	.018	.049

Advantages of Diversion

- ▶ When you were taught elasticities you were taught they were a **unit free** comparison across products, markets, etc.
- ▶ But that is only true of **own elasticities** not cross elasticities.
- ▶ Is $\epsilon_{jt} = .01$ or $\epsilon_{jt} = .03$ a better substitute? We can't tell.
 - You need to take $\epsilon_{jk} \cdot \sigma_k$ to know. If you take $\epsilon_{jk} \cdot \frac{\sigma_k}{p_j} = D_{jk}$ (back at diversion).
 - Because it tells us **fraction of switchers** choosing k we can compare across settings (since share sums to one).
 - ie: diversion ratio $D_{jk} = 0.1$ actually tells me something!
- ▶ Whenever you are tempted to report **cross elasticities** report **diversion ratio** instead.

	Cheerios	Special K	Corn Flakes	Reese's Puffs	Capt Crunch	Froot Loops	Shares
HN Cheerios	5.07	4.27	3.75	5.33	3.58	3.48	2.69
Frosted Flakes	2.46	2.54	4.54	4.00	5.35	7.24	2.65
Cheerios	-2.46	5.91	3.13	3.19	1.36	1.77	2.10
Honey Bunches	2.47	2.51	2.21	2.08	1.94	1.99	1.47
Cinn Toast Crunch	3.43	2.10	1.69	3.00	1.78	1.84	1.43
Froot Loops	1.26	1.19	1.64	1.69	1.82	-2.71	1.18
Lucky Charms	2.18	1.64	1.57	2.99	1.59	1.58	1.14
Frosted Mini-Wheats	0.36	0.50	0.74	0.68	0.87	1.27	1.01
Corn Flakes	2.01	2.18	-2.64	1.31	1.24	1.52	0.98
Rice Krispies	1.50	1.72	1.56	0.89	0.68	1.25	0.96
Apple Jacks	0.91	0.80	1.24	1.27	1.42	2.45	0.85
Raisin Bran (KEL)	0.46	0.47	0.63	0.78	0.82	1.24	0.79
Special K Red Berry	0.96	1.45	0.95	0.78	0.68	0.90	0.75
Special K	2.06	-2.66	1.18	0.71	0.44	0.58	0.74
MG Cheerios	1.11	0.99	0.75	0.89	0.54	0.66	0.71
Reese's Puffs	1.36	0.86	0.87	-2.70	1.08	1.01	0.69
Life	1.15	1.12	1.05	1.02	1.72	0.89	0.68
Cocoa Puffs	1.18	0.92	0.95	1.47	1.05	0.97	0.67
Capt Crunch	0.63	0.58	0.88	1.21	-2.68	1.19	0.62
Capt Crunch Berry	0.68	0.61	0.83	1.15	3.29	1.00	0.58
Corn Pops	0.43	0.43	0.71	0.66	0.75	1.45	0.56
Cinn Life	0.76	0.75	0.83	0.84	1.59	0.78	0.54
Fruity Pebbles	0.61	0.59	0.71	0.71	0.75	0.77	0.44

Where do Diversion Ratios come from?

(Stolen from Conlon and Mortimer (2021))

The diversion ratio $D_{jk}(p_j, x) \equiv \frac{\partial q_k}{\partial p_j}(p_j, x) / -\frac{\partial q_j}{\partial p_j}(p_j, x)$ can be obtained as the limit of the Wald estimator where the price increase (or decrease) becomes small, so long as demand slopes *strictly* downwards $\frac{\partial q_j}{\partial p_j} < 0$.

$$\lim_{p'_j \rightarrow p_j} \frac{q_k(p'_j, x) - q_k(p_j, x)}{-(q_j(p'_j, x) - q_j(p_j, x))} \rightarrow \frac{\frac{\partial q_k}{\partial p_j}(p_j, x)}{-\frac{\partial q_j}{\partial p_j}(p_j, x)} \equiv D_{jk}(p_j, x)$$

Theorem (Conlon Mortimer (2021))

Under the following conditions:

- (a) Mutually Exclusive and Exhaustive Discrete Choice: $d_{ij} \in \{0, 1\}$ and $\sum_{j \in \mathcal{J}} d_{ij} = 1$.*
- (b) Exclusion: $u_{ik}(p_j, x) = u_{ik}(p'_j, x)$ for all $k \neq j$ and any (p_j, p'_j) ;*
- (c) Monotonicity: $u_{ij}(p'_j, x) \leq u_{ij}(p_j, x)$ for all i and any $(p'_j > p_j)$; and*
- (d) Existence of a first-stage: $\mathbb{P}(d_{ij}(p_j, x) = 0) \neq \mathbb{P}(d_{ij}(p'_j, x) = 0)$ for $(p'_j > p_j)$;*
- (e) Random Assignment: $(u_{ij}(p_j, x), u_{ik}(p_j, x)) \perp p_j$.*

then the Wald estimator:

$$\frac{q_k(p'_j, x) - q_k(p_j, x)}{q_j(p'_j, x) - q_j(p_j, x)} = \mathbb{E}[D_{jk,i}(x) | d_{ij}(p_j, x) > d_{ij}(p'_j, x)]$$

Outcome $Y_i \in \{0, 1\}$ denotes the event that consumer i purchases product k : $d_{ik}(p_j) = 1$.

Treatment $T_i \in \{0, 1\}$ denotes the event that consumer i does **not** purchase product j . In other words $T_i = 0$ implies $d_{ij}(p_j) = 1$ and $T_i = 1$ implies $d_{ij}(p_j) = 0$.

Instrument $Z_i = p_j$ the price of j induces consumers into not purchasing j .

What can we learn from a change in p_j ?

$$\begin{aligned}
 \text{Wald}(p_j, p'_j, x) &= \int_{p_j}^{p'_j} D_{jk}(p_s, x) w(p_s) \partial p_s \text{ with } w(p_s) = \frac{\frac{\partial q_j(p_s, x)}{\partial p_j}}{\int_{p_j}^{p'_j} \frac{\partial q_j(p_t, x)}{\partial p_j} \partial p_t} \\
 &= \int_{p_j}^{p'_j} \int D_{jk,i}(p_s, x) w_i(p_s, x) \partial p_s \partial F_i \text{ with } w_i(p_s, x) = \frac{\left| \frac{\partial q_{ij}(p_s, x)}{\partial p_j} \right|}{q_j(p_j, x) - q_j(p'_j, x)} \\
 &= \int D_{jk,i}(x) w_i(z_j, z'_j, x) \partial F_i \text{ with } w_i(z_j, z'_j, x) = \frac{q_{ij}(z_j, x) - q_{ij}(z'_j, x)}{q_j(z_j, x) - q_j(z'_j, x)}
 \end{aligned}$$

- Individual diversion ratios in logit family are $D_{jk,i} = \frac{\sigma_{ik}}{1 - \sigma_{ij}}$ and don't depend on p_j, z_j
- Which individuals respond to z_j or p_j determines the weighting scheme only.

Again $D_{jk,i} = \frac{\sigma_{ik}}{1-\sigma_{ij}}$ for logit family:

	$w_{ij}(x) \propto$
second choice data	$\sigma_{ij}(x)$
price change $\frac{\partial}{\partial p_j}$	$\sigma_{ij}(x) \cdot (1 - \sigma_{ij}(x)) \cdot \alpha_i $
characteristic change $\frac{\partial}{\partial x_j}$	$\sigma_{ij}(x) \cdot (1 - \sigma_{ij}(x)) \cdot \beta_i $
small quality change $\frac{\partial}{\partial \xi_j}$	$\sigma_{ij}(x) \cdot (1 - \sigma_{ij}(x))$
finite change in z_j	$\sigma_{ij}(z'_j) - \sigma_{ij}(z_j)$

- Normalize so that above weights sum to one $w_{ij}^* = w_{ij} / \int w_{ij} dF_i$
- For plain logit $D_{jk,i} = \frac{\sigma_k}{1-\sigma_j}$ for all i and weights don't matter!

Here we get a really simple decomposition:

$$D_{j \rightarrow k} = \sum_{i \in \mathcal{I}} \underbrace{\frac{\sigma_{ik}}{1 - \sigma_{ij}}}_{D_{jk,i}} \cdot \underbrace{\frac{w_i \cdot \sigma_{ij}}{\sigma_j}}_{\gamma_{ij}}$$

Idea: different interventions change **weights** but not **individual diversion ratios**:

- **Individual Diversion Ratios** $D_{jk,i} : \mathbb{P}(i \text{ chooses } k \mid \text{ doesn't choose } j)$.
- **Weights** $\gamma_{ij} : \text{For second choices: } \mathbb{P}(\text{Type } i \mid \text{ leaves } j) = \mathbb{P}(\text{Type } i \mid \text{ chooses } j)$.

For infinitesimal price changes $\gamma_{ij} = \frac{w_i \cdot |\beta_i^p| \cdot \sigma_{ij}(1 - \sigma_{ij})}{\sum_{i' \in \mathcal{I}} w_{i'} \cdot |\beta_{i'}^p| \cdot \sigma_{i'j} \cdot (1 - \sigma_{i'j})}$ or finite: $\gamma_{ij}(p_j, p'_j) = \frac{\sigma_{ij}(p'_j) - \sigma_{ij}(p_j)}{\sigma_j(p'_j) - \sigma_j(p_j)}$.
(Types that are: price sensitive and likely to choose j).

But we could do this with anything (not just price!)

Application of CM Decomposition: Customer Overlap (Einav, Guido, Klenow, AER:I 2026)

Panel data (credit cards, cell-phone traces, Nielsen panelists) record T_i purchases per consumer but no second-choice question. Let q_{ik} be i 's purchases of k , $\sum_k q_{ik} = T_i$, and $\mathcal{C}_j = \{i : q_{ij} \geq 1\}$ the consumers **ever observed buying j** (Starbucks). EGK propose the **customer overlap**:

$$C_{j \rightarrow k} = \frac{\sum_{i \in \mathcal{C}_j} q_{ik}}{\sum_{i \in \mathcal{C}_j} (T_i - q_{ij})} = \sum_{i \in \mathcal{C}_j} \underbrace{\frac{\hat{\sigma}_{ik}}{1 - \hat{\sigma}_{ij}}}_{\hat{D}_{jk,i}} \cdot \underbrace{\frac{T_i - q_{ij}}{\sum_{i' \in \mathcal{C}_j} (T_{i'} - q_{i'j})}}_{w_i^{\mathcal{C}}} \quad \text{with } \hat{\sigma}_{ij} = q_{ij}/T_i.$$

- Fits the CM decomposition: a weighted average of **individual diversion ratios**, weights $\propto$ non- j purchases.
- Two forces pull in opposite directions. *Conditional* on $i \in \mathcal{C}_j$, weight is heaviest on consumers **least** loyal to j ; *selection* into $\mathcal{C}_j$ favors the **most** loyal (they are the ones observed buying j at all).
- How much they cancel depends on panel length T_i . Assume mixed logit with persistent V_{ij} and iid ε_{ijt} across purchase occasions, so $q_{ij} \sim \text{Binomial}(T_i, \sigma_{ij})$ and we can work it out (next slide).

What Does Overlap Measure? (Conlon, Mortimer, Sarkis 2026, Appendix)

Population analogue (type weights w_i , per-type panel weight $\bar{w}_i(\sigma_{ij}) \equiv \mathbb{E} [T_i (1 - (1 - \sigma_{ij})^{T_i-1}) \mid i]$):

$$C_{j \rightarrow k} = \frac{\sum_i w_i \sigma_{ik} \bar{w}_i(\sigma_{ij})}{\sum_i w_i (1 - \sigma_{ij}) \bar{w}_i(\sigma_{ij})} = \sum_i \underbrace{\frac{w_i (1 - \sigma_{ij}) \bar{w}_i(\sigma_{ij})}{\sum_{i'} w_{i'} (1 - \sigma_{i'j}) \bar{w}_{i'}(\sigma_{i'j})}}_{\gamma_{ij}^{(C)}} \cdot D_{jk,i}$$

Panel length	$\gamma_{ij}^{(C)} \propto$	$C_{j \rightarrow k}$ equals
$T_i = 2$ for everyone	$w_i \sigma_{ij} (1 - \sigma_{ij})$	quality diversion $D_{j \rightarrow k}^\xi$ (the $\partial/\partial \xi_j$ row of the weights table), exactly
small σ_{ij} , moderate T_i	$\approx w_i \sigma_{ij} (1 - \sigma_{ij})$	$\approx$ quality diversion (degrades for loyal j -buyers)
$T_i \rightarrow \infty$	$w_i (1 - \sigma_{ij})$	IIA logit $\sigma_k/(1 - \sigma_j)$: all heterogeneity gone

- The longer the panel, the **less** the raw overlap tells you: every type eventually enters $\mathcal{C}_j$ and the selection no longer discriminates.
- Regimes assume T_i independent of type; heavy users observed longer tilt the weights by $\mathbb{E}[T_i \mid i]$.

A Better Overlap Measure: Odds-Ratio Reweighting (CMS 2026, Appendix)

Same data (q_{ij}, q_{ik}, T_i) . Weight each consumer by their share of j 's purchases q_{ij}/Q_j instead of by their non- j purchases (equivalently: reweight EGK by the odds $\hat{\sigma}_{ij}/(1 - \hat{\sigma}_{ij})$):

$$A_{j \rightarrow k} = \frac{1}{Q_j} \sum_{i: q_{ij} < T_i} \frac{q_{ij} q_{ik}}{T_i - q_{ij}} = \sum_i \underbrace{\frac{\hat{\sigma}_{ik}}{1 - \hat{\sigma}_{ij}}}_{\hat{D}_{jk,i}} \cdot \underbrace{\frac{\hat{\sigma}_{ij}}{\sum_{i'} \hat{\sigma}_{i'j}}}_{\hat{\gamma}_{ij}^{\text{second choice}}}, \quad Q_j = \sum_i q_{ij}.$$

- These are the empirical **second-choice weights** $\gamma_{ij} = w_i \sigma_{ij} / \sigma_j$ from two slides ago. No conditioning on $\mathcal{C}_j$; consumers with $q_{ij} = 0$ or $q_{ij} = T_i$ contribute nothing.
- Population limits: $T_i = 2 \Rightarrow$ quality diversion (same as EGK); $T_i \rightarrow \infty \Rightarrow$ **second-choice diversion** $D_{j \rightarrow k}$. It **improves** with panel length rather than collapsing to IIA.
- Either version is a legitimate input to the product-space estimator in the next deck, as long as the model-side formula matches (quality vs. second-choice weights differ by a factor $(1 - \sigma_{ij})$, ≈ 1 when shares are small).
- Caveats: $\hat{\sigma}_{ij}$ is noisy at small T_i ; assumes a stable environment over the panel (no entry/exit, price changes) and persistent V_{ij} with iid $\varepsilon_{ij,t}$.

Welfare and WTP: Diversion to the Outside Good

Recall the worked removal example from the logit lecture: for a member of the logit family, the welfare loss from removing j is $\Delta CS_i = \frac{1}{\alpha_i} \log(1 - \sigma_{ij})$. The same object, written in the language of this lecture:

$$WTP_i(j) = \log \left(\frac{\sigma_{i0}(\mathcal{J} \setminus j)}{\sigma_{i0}(\mathcal{J})} \right) = \log \left(1 + \frac{D_{j0,i} \sigma_{ij}}{\sigma_{i0}(\mathcal{J})} \right) = \log \left(1 + \frac{\sigma_{ij}}{1 - \sigma_{ij}} \right) \approx \frac{\sigma_{ij}}{1 - \sigma_{ij}}$$

using $\sigma_{i0}(\mathcal{J} \setminus j) = \sigma_{i0}(\mathcal{J}) + D_{j0,i} \cdot \sigma_{ij}$ and $D_{j0,i} = \frac{\sigma_{i0}}{1 - \sigma_{ij}}$.

- ▶ Welfare and willingness to pay are mostly about **diversion to the outside good** (HW2).
- ▶ Individual willingness to pay is exclusively about **own share**, not the presence of close substitutes. Is this a good property or not?
- ▶ This is the $WTP_i(j)$ that drives the hospital-network literature (Capps, Dranove, Satterthwaite 2003), coming up in the Health IO lecture.

Practical Tips: Checklist

- ▶ How much diversion to outside good?
- ▶ What are top substitutes by diversion? How much do they get?
- ▶ Do identities of best substitutes differ across products?
- ▶ How much diversion goes to own products vs. competitors?
- ▶ For mergers, what would **divestiture** candidates be?